Clinton H. Durney

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EDUCATION

The University of British Columbia, Vancouver, BC, Canada

Ph.D. Applied Mathematics 2020
Dissertation: *Applications of vertex modelling to epithelial morphogenesis*
Advisor: Prof. James J. Feng

The Ohio State University, Columbus, OH, USA

M.S. Mathematics 2013
Thesis: *A two-component model for bacterial chemotaxis*

Virginia Polytechnic Institute and State University (Virginia Tech), Blacksburg, VA, USA

B.S. Physics; B.S. Mathematics 2011

RESEARCH EXPERIENCE

BC Cancer Research Institute, Vancouver, BC, Canada

Research Methodologist, Population Health Sciences 2023–

University of Illinois Chicago (UIC), Chicago, IL, USA

Consultant, Department of Medicine 2024–

University of Michigan, Ann Arbor, MI, USA

Affiliate Scientist, School of Public Health, Epidemiology Department 2023–

John Innes Centre, Norwich, UK

Visiting Research Scientist, Computational and Systems Biology 2023–2024
Postdoctoral Scientist, Computational and Systems Biology 2020–2023

The University of British Columbia, Vancouver, BC, Canada

Graduate Research Assistant, Department of Mathematics 2015–2020

The Ohio State University, Columbus, OH, USA

Graduate Research Assistant, Department of Mathematics 2011–2013

Virginia Tech, Blacksburg, VA, USA

Research Assistant, Department of Physics 2010–2011

RESEARCH SUPPORT

- **Next Generation of Scientists Award**, Cancer Research Society 2026–2029

  Role: Principal Investigator
  Phase I fellowship (2026–2027): *Advancing Air Pollution Exposure Assessment and Establishing the Pre-Nodule Imaging Gap in Lung Cancer Screening*
  Phase II operating grant (2027–2029): *Radiomic Trajectory Analysis of $PM_{2.5}$-Driven Parenchymal and Airway Change in Longitudinal Lung Cancer Screening Cohorts*
  Amount: $190,000 CAD total ($50,000 Phase I; $140,000 Phase II)

- **Pilot Project Award** 2024–2025

  Center for the Assessment of Tobacco Regulations (CAsToR), University of Michigan
  Role: Principal Investigator
  Understanding the Role of Social Network Dynamics on Adolescent Tobacco and Nicotine Use

Funded under NIH/FDA 2U54CA229974
Amount: $40,000 USD

PUBLICATIONS (* indicates equally contributing authors, † indicates student or mentee under direct supervision)

17. **Durney, C.H.**, Xu, K., Atkar-Khattra, S., Lam, S., and Meza, R. Temporal dynamics of Sybil AI lung cancer risk scores in longitudinal screening cohorts. *In Preparation (2026)*

16. **Durney, C.H.**, Lam, S., Chiu, C.H., *et al.*, and Meza, R. Ambient air pollution exposure correlates with lung cancer screening outcomes among never smokers in the Taiwan TALENT study. *Submitted, JTO Clinical and Research Reports (2026)*

15. Wang, P., Naso, J.R., **Durney, C.H.**, *et al.*, and Lockwood, W.W. $PM_{2.5}$ exposure induces cancer initiation and a distinct mutational signature in lung adenocarcinoma patients who never smoked. *Submitted, Clinical Cancer Research (2026)*

14. Pasquinelli, M.M., **Durney, C.H.**, *et al.*, and Weinberg, F. Validation of the Sybil future lung cancer risk prediction model in a predominantly Black screening cohort. *Journal of Thoracic Oncology, in press (2026)*. doi:10.1016/j.jtho.2026.106128

13. Fung, L., Joseph, B., Lo, T., *et al.*, (including **Durney, C.H.**, †Tawara, A.), and Lim, E. Clinical and molecular correlates of digitally quantified anthracosis in lung tissue slides: Utility of an automated quantitative workflow. *The Journal of Pathology: Clinical Research* 12(4):e70110 (2026)

12. Pandya, T., Zagorulya, M., Leung, M., *et al.*, (including **Durney, C.H.**), and Swanton, C. Plasma signals of lung tumor promotion for molecular cancer prevention. *Cell* 189:3903–3921.e26 (2026)

    Press: The New York Times, June 4, 2026

11. **Durney, C.H.**, †Tawara, A., Brauer, M., Atkar-Khattra, S., Myers, R., Meza, R., and Lam, S. APEX: A web-based tool for assessing long-term outdoor $PM_{2.5}$ exposure—Brief report. *JTO Clinical and Research Reports* 7(5):100975 (2026)

10. **Durney, C.H.**, Meza, R., Jimenez-Mendoza, E., Xu, K., Levy, D.T., and Friedman, A. Recent trends in sales of electronic nicotine devices and oral nicotine pouches by flavour. *Tobacco Control*, online ahead of print (2025). doi:10.1136/tc-2025-059478

9. Wilson, M.J., McGregor, S., **Durney, C.H.**, Tomkins, M., Armand, J., Smith, R.S., Gray, J.E., Morris, R.J., and Fleming, A.J. Symplastic guard cell connections buffer pressure fluctuations to promote stomatal function in grasses. *New Phytologist* 246(1) (2025)

8. ***Durney, C.H.**, *Wilson, M.J., McGregor, S., Armand, J., Smith, R.S., Gray, J.E., Morris, R.J., and Fleming, A.J. Grasses exploit geometry to achieve improved guard cell dynamics. *Current Biology* 33(13) (2023)

7. Ashour, D.J., **Durney, C.H.**, Planelles-Herrero, V.J., Stevens, T.J., Feng, J.J., and Röper, K. Zasp52 strengthens whole embryo tissue integrity through supracellular actomyosin networks. *Development* 150(7):dev201238 (2023)

6. Carroll, S., Amsbury, S., **Durney, C.H.**, Smith, R.S., Morris, R.J., Gray, J.E., and Fleming, A.J. Altering arabinans increases *Arabidopsis* guard cell flexibility and stomatal opening. *Current Biology* 32(14) (2022)

5. **Durney, C.H.**, and Feng, J.J. A three-dimensional vertex model for *Drosophila* salivary gland invagination. *Physical Biology* 18(4):046005 (2021)

4. **Durney, C.H.**, Harris, T.J.C., and Feng, J.J. Dynamics of PAR proteins explain the oscillation and ratcheting mechanisms in dorsal closure. *Biophysical Journal* 115(11) (2018)

3. **Durney, C.H.**, Case, S.O., Pleimling, M., and Zia, R.K.P. Stochastic evolution of four species in cyclic competition. *Journal of Statistical Mechanics: Theory and Experiment* 2012(6) (2012)

2. **Durney, C.H.**, Case, S.O., Pleimling, M., and Zia, R.K.P. Saddles, arrows, and spirals: Deterministic trajectories in cyclic competition of four species. *Physical Review E* 83(5) (2011)

1. Case, S.O., **Durney, C.H.**, Pleimling, M., and Zia, R.K.P. Cyclic competition of four species: Mean-field theory and stochastic evolution. *EPL (Europhysics Letters)* 92(5) (2010)

SELECTED PRESENTATIONS (* indicates invited)

- *Partial flavour bans displace rather than reduce nicotine consumption: Evidence from US retail sales data*, 2026 World Conference on Lung Cancer, Seoul, South Korea, 2026

- **Decoding Sybil: How nodule morphology shapes an AI lung cancer risk score across serial CT scans*, UBC Radiology Clinical & Fundamental Science Research Day, Vancouver, BC, Canada, 2026
- **Mathematical models of tobacco use: Products, flavors, and networks*, UBC Mathematical Biology Seminar, Vancouver, BC, Canada, 2025
- **Analysis of longitudinal low-dose CT Sybil scores in two Canadian cohorts*, 2025 World Conference on Lung Cancer, Barcelona, Spain, 2025
- *Recent trends in nicotine product use by flavor*, Society for Research on Nicotine and Tobacco (SRNT) Annual Meeting, New Orleans, LA, USA, 2025
- **Practical implementation of outdoor air pollution exposure measurement for lung cancer risk assessment*, 2024 World Conference on Lung Cancer, San Diego, CA, USA, 2024
- **Grasses exploit geometry to achieve improved guard cell dynamics*, John Innes Centre–The Sainsbury Laboratory Annual Science Meeting, Norwich, UK, 2023
- **The mechanics of biological valves – how plants regulate photosynthesis in grasses*, UBC Botany Seminar, Vancouver, BC, Canada, 2023
- *An imaging and FEM study into the mechanics of biological valves – how plants regulate photosynthesis in grasses*, Physics of Life, Harrogate, UK, 2023
- **Revisiting the "mechanical advantage" of subsidiary cells in grass stomata*, BIRS-CMO, Oaxaca, Mexico (hybrid), 2022
- *Shape-shifting stomata: Mechanical interactions of grass stomata*, Plant Biomechanics Conference, Lyon, France, 2022
- *Mechanical interactions of graminoid (grass) stomata*, Computational and Systems Biology Seminar, John Innes Centre, Norwich, UK, 2022
- **Quantifying cellular contributions to* Drosophila *salivary gland invagination*, Society for Mathematical Biology (SMB) Annual Meeting, Virtual, 2020
- *3D modelling of* Drosophila *salivary gland invagination*, UBC Mathematical Biology Seminar, Vancouver, BC, Canada, 2020
- *Dynamics of PAR proteins explain oscillation and ratcheting mechanisms in* Drosophila *dorsal closure*, Society for Mathematical Biology (SMB) Annual Meeting, Montréal, QC, Canada, 2019
- *Dynamics of PAR proteins explain oscillation and ratcheting mechanisms in* Drosophila *dorsal closure*, Frontiers in Biophysics, Simon Fraser University, Vancouver, BC, Canada, 2019
- *Translocation and interaction of PAR proteins explain oscillation and ratcheting mechanisms during* Drosophila *dorsal closure*, APS March Meeting, Los Angeles, CA, USA, 2018
- *A proposed mechanochemical process for* Drosophila *dorsal closure*, UBC Mathematical Biology Seminar, Vancouver, BC, Canada, 2017
- *Four species in cyclic competition: mean-field and stochastic results*, Workshop for Young Researchers in Mathematical Biology, Mathematical Biosciences Institute, Columbus, OH, USA, 2011
- *Mean-field theory of four species in cyclic competition*, APS March Meeting, Dallas, TX, USA, 2011
- *Mean-field theory (MFT) predictions for four species in cyclic competition*, 104th Statistical Mechanics Conference, Rutgers University, New Brunswick, NJ, USA, 2010

RESEARCH TOOLS AND SOFTWARE

- **A**ir **P**ollution **Ex**posure Tool (APEX), Creator and Lead Developer
  - Web-based tool for estimating long-term $PM_{2.5}$ exposure from residential history
  - Used by clinicians and in epidemiological studies by researchers worldwide
  - Available at apex.bccrc.ca

STUDENTS MENTORED

- Caleb Cheuk, UBC Computer Science, NSERC USRA Undergraduate Researcher, Summer 2025

- Aadetri Tawara, UBC Computer Science and Statistics, Undergraduate Researcher, Summer 2025

TEACHING EXPERIENCE

The University of British Columbia, Department of Mathematics, Vancouver, BC, Canada

Graduate Teaching Assistant 2015–2020
· Facilitated and taught workshops in calculus, multivariable calculus, vector calculus, differential equations, and mathematical computing (Python and MATLAB)
· Developed and coded WeBWorK questions for applied linear algebra

MATH 599 Mathematics Teaching Techniques 2018
Participant in graduate-level course specialized for teaching mathematics at the post-secondary level

Instructional Skills Workshop 2017
Certificate (accredited by UBC) obtained through participating in workshop on teaching practice, theory application, and topical sessions aimed at those teaching mathematics

Varee International School, Chiang Mai, Thailand

Instructor, A and AS Level Physics and Mathematics 2013–2015
· Taught to the Cambridge International Examinations (CIE) qualifications
· Taught Pure Mathematics 1–3, Statistics 1, and Mechanics 1 syllabi
· Managed a multicultural and internationally diverse classroom

The Ohio State University, Department of Mathematics, Columbus, OH, USA

Graduate Teaching Assistant 2011–2013
· Recitation leader for calculus, business calculus, and mathematical modeling for the life sciences
· Tutor in the Mathematics Learning Center

PROFESSIONAL TRAINING

- Regression Modeling Strategies Short Course, Vanderbilt University, 2025
- Simulation Modeling of Tobacco Use, Summer Session in Epidemiology, University of Michigan, 2024
- Multiscale Modeling of Plant Growth, Pattern Formation and Actuation, BIRS Casa-Matemática Oaxaca (BIRS-CMO) Workshop, 2022
- Early Career Researchers Peer Review Program, Nature Publishing Group, 2021
- Bridging Cellular and Tissue Dynamics from Normal Development to Cancer: Mathematical, Computational, and Experimental Approaches, BIRS Workshop, 2019
- Stochastic and Deterministic Modelling with PDEs, Pacific Institute for the Mathematical Sciences (PIMS) Workshop, 2017
- Stochastics Applied to Biological Systems, Mathematical Biosciences Institute (MBI-NIMBioS-CAMBAM) Workshop, 2012
- Evolution Equations: A Workshop in honor of Terence Tao, Northwestern University, 2012
- SAMSI Undergraduate Workshop, The Statistical and Applied Mathematical Sciences Institute (SAMSI), 2011

REVIEWER

- *Grant review:* Canada Research Chairs
- *Journals:* Cancer Medicine, PLOS Computational Biology, PLOS ONE, Biophysical Journal, The Plant Journal, Quantitative Plant Biology, Bioinspiration & Biomimetics

AWARDS

- Next Generation of Scientists Award, Cancer Research Society, 2026
- Stanley M. Grant Scholarship in Mathematics, UBC, 2019
- International Tuition Award, UBC, 2015–2020

- Faculty of Science PhD Tuition Award, UBC, 2015–2020
- Daniel C. and Delia F. Grand Scholarship, Virginia Tech, 2009
- Richard C. Coleman Scholarship, Virginia Tech, 2008

PROFESSIONAL MEMBERSHIP

- Society for Research on Nicotine and Tobacco
- International Association for the Study of Lung Cancer (IASLC)